

Anshuman Goel

+91 9056953665 | anshg112005@gmail.com | LinkedIn | GitHub |

EDUCATION

Thapar Institute of Engineering and Technology, Patiala
B.E. Computer Science CGPA:7.3

Patiala, India
Aug 2023 - May 2027 (Expected)

Guru Nanak Foundation Public School, Patiala
CBSE Board, Class 12th

Patiala, India
Apr 2021 - Mar 2023

PROJECTS

Student Verification Portal | *React, TypeScript, Supabase, PostgreSQL*

2026

- *Built VerifyHub*, a full-stack verification platform with a real-time WebSocket leaderboard, automating validation of **6,000+** student submissions and reducing manual verification effort by over **90%**.
- *Developed serverless Deno Edge Functions* with parallel async scraping and a custom Jaccard set-intersection fuzzy name-matching algorithm, achieving automated identity resolution across Coursera and LinkedIn.
- *Secured backend against SSRF, IP spoofing, and JWT replay attacks* via strict hostname allowlisting, user ID-based rate limiting, and idempotent DB update logic in a TypeScript/Supabase stack.

Distributed In-Memory Cache | *C++, REST API*

2026

- Implemented consistent hashing with virtual nodes; reduced load imbalance **85%** (7% → 1%) across 10,000 keys by scaling from 10 to 500 virtual nodes.
- Integrated multi-threaded concurrent access using reader-writer locks and atomic counters, achieving **682,000+** ops/sec under 8-thread load with zero data races.
- Simulated node failure mid-operation with automatic key redistribution (3,312 keys redistributed); maintained 1% imbalance and zero data loss across surviving nodes. Exposed live REST API via cpp-http lib (/get, /put, /fail, /report).
- Applied SOLID principles by modularizing hashing (strategy pattern via IHasher), separating concerns (HashRing, CacheNode, DistributedCache), and enabling extensibility for alternate hashing strategies.

WORK EXPERIENCE

Software Engineering Intern | *SafeSpace*

May 2025 – July 2025

- Developed a multimodal stress detection system (CNN, DNN, TensorFlow, PyTorch) fusing physiological signals, facial expressions, audio, and survey data - Achieving **94.1%** accuracy on RAVDESS, outperforming **4** prior state-of-the-art baselines.
- Designed a novel Agreement-Aware Fusion (AAF) algorithm for dynamic, instance-level weighting of 4 deep learning model outputs, improving classification robustness over standard ensemble methods without any learned parameters.
- Integrated a lightweight LLM (Phi-2) with OpenAI Whisper for real-time speech transcription and personalized mental health recommendations, deployed end-to-end as a web and mobile application.

SKILLS SUMMARY

- **Languages:** C++, Python, JavaScript (TypeScript), SQL
- **Libraries/Frameworks:** React, Node.js, Express.js, Flask, Tailwind CSS, React Router, React Query, TensorFlow, Scikit-learn, NumPy, Pandas
- **Tools & Platforms:** Supabase (PostgreSQL), MySQL, MongoDB, Upstash Redis, AWS (Basic), Docker, Git, GitHub, Vercel, Postman, Vite
- **Concepts:** Data Structures & Algorithms, Operating System, System Design, DBMS, Machine Learning, Deep Learning, OOPs,

ACHIEVEMENTS

Innovation Award – Cyber AI Hackathon 2025

Nov 2025

University of Derby (UK) & American Society for Engineers

Project: MediMind

- Recognized for building an AI-driven pharmacy intelligence platform enabling early epidemic detection and smart inventory management using real-world, imperfect data.

LEADERSHIP & EXTRACURRICULARS

Head of Logistics – Mudra Cultural Society, Thapar Institute

Jan 2024 – Present

- Managed logistics for flagship college events with **1000+** attendees, coordinating vendors and team operations.
- Led event planning committee and optimized resource allocation resulting in **20%** cost reduction.

Head of Logistics – MUN Society, Thapar Institute

Aug 2023 – Present

- Coordinated MUN simulations for **300+** delegates, managing schedules, venue arrangements, and delegate kits.
- Streamlined registration process and improved participant satisfaction scores by **25%**.